

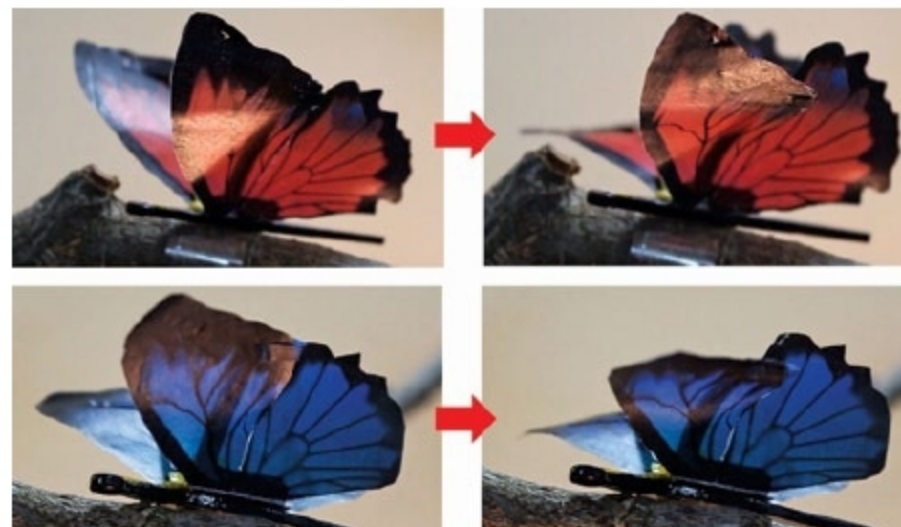
🔔 Researchers develop an ultrathin, artificial muscle for soft robotics

Researchers at Korea Advanced Institute of Science and Technology (KAIST) in South Korea have developed an ultrathin artificial muscle, also called an actuator, for soft robotics. They demonstrated the new technology with a robotic blooming flower brooch, dancing robotic butterflies and fluttering leaves on a kinetic work of art.

The robotic equivalent of a muscle that can move is called an actuator. The actuator expands, contracts or rotates like muscle fibres using a stimulus such as electricity. Engineers around the world are striving to develop more dynamic actuators that respond quickly, can bend without breaking and are very durable. Soft, robotic muscles could have a wide variety of applications, from wearable electronics to advanced prosthetics.

The actuator looks like a skinny strip of paper a few centimetres long. Researchers used a particular type of material called MXene, which is a class of compounds that have layers only a few atoms thick. MXene material (T₃C₂T_x) is made of thin layers of titanium and carbon compounds. It is not flexible by itself—sheets of material flake off the actuator when bent

in a loop. But when MXene is ionically cross-linked—connected through an ionic bond—to a synthetic polymer, the combination of materials makes the actuator flexible, while still maintaining strength and conductivity, which is critical for movements driven by electricity.



Artificial muscles bloom, dance and wave (Credit: KAIST)

🔔 Failed cancer drug being repurposed as organic semiconductors

Many potential pharmaceuticals end up failing during clinical trials, but thanks to new research from University of Illinois, biological molecules once considered for cancer treatment are now being repurposed as organic semiconductors for use in chemical sensors and transistors.

Organic semiconductors are responsible for things like flexible electronics and transparent solar cells, but researchers are working to expand their use in biomedicine and devices that require interaction between electrically-active and biological molecules.

Chemical and biomolecular engineering professor Ying Diao says that she was surprised when the two avenues of her research, namely, pharmaceutical development and printable

electronics, merged in her lab with the discovery of semiconductor-like features in a well-studied bioactive molecule. The molecule, which inserts itself into DNA to prevent replication, was once explored as a potential anti-cancer agent.

These molecules, called DNA topoisomerase inhibitors, are flat and contain neatly-stacked columns of electrically-conductive molecular rings—features that make a good semiconductor. Distinct from a typical semiconductor, these molecular columns are linked together by hydrogen bonds that can move charges from column to column, forming bridges that transform the entire molecular assembly into a semiconductor, she adds.

🔔 Study says humans will listen to virtual agents except when the going gets tough

We may listen to facts from Siri or Alexa, or get directions from Google Maps or Waze, but would we let a virtual agent enabled by artificial intelligence (AI) help mediate conflict among team members? A new study says, not just yet.

Researchers from University of Southern California (USC) and University of Denver have created a simulation in which a three-person team was supported by a virtual agent *avatar* on screen in a mission that was designed to ensure failure and elicit conflict. The study was designed to look at virtual agents as potential mediators to improve team collaboration during conflict mediation.

While some researchers (Gale Lucas and Jonathan Gratch of USC Viterbi School Engineering and USC Institute for Creative Technologies who contributed to this study) had previously found that one-on-one human interactions with a virtual agent therapist yielded more confessions, in this study 'Conflict

Mediation in Human-Machine Teaming: Using a Virtual Agent to Support Mission Planning and Debriefing,' team members were less likely to engage with a male virtual agent named Chris when conflict arose.

Participating members of the team did not physically accost the device, but rather were less engaged and less likely to listen to the virtual agent's input once failure ensued and conflict arose among team members.

The study was conducted in a military academy environment in which 27 scenarios were engineered to test how the team that included a virtual agent would react to failure and the ensuing conflict. The study found that the team did respond socially to the virtual agent during planning of the mission it was assigned (nodding, smiling and recognising the virtual agent's input by thanking it), but the longer the exercise progressed, engagement with the virtual agent decreased.